# Solutions and teaching notes, chapter 7

**Teacher use.** The following answers concern the tasks not solved in the lesson. First have pupils make their approach explicit; provide graduated help before the solution. The durations are estimates to be calibrated in class.

## Diagnostic and remediation

1. 0. *Action:* Go back to the orthogonality criterion.

2. Yes: $2 \quad -2 = 0$. *Action:* Substitute x and y simultaneously.

3. $x = 3,\ y = 2$. *Action:* Add, then substitute.

4. 5. *Action:* Calculate the vector and then its norm.

5. $(x-2)^2 - 4$. *Action:* Expand to check.

6. The only pair (0; 0). *Action:* Use the sum of two non-negative squares.

## Investigation

Trials suggest H(2;4), but Pythagoras' theorem in PHM proves minimality. The projection is calculated with the normal (2;-1).

## Direct applications

1. $3(x-2) + 4(y+1) = 0$, i.e. $3 \quad x + 4y - 2 = 0$.

2. $H(2;\ 1)$ and distance 2; the normal is horizontal.

3. $(x+1)^2 + (y-4)^2 = 0$ : the single point $(\ -1;\ 4)$, not a circle of strictly positive radius.

4. $(x-1)^2 = 0$, so the only point is (1; 1); the line is tangent to the circle.

## Exercises: methods and answers

**Exercise 7.1 - Constructing a line.** $2(x+1) + 5(y-3) = 0$, i.e. $2 \quad x + 5y - 13 = 0$. Direction vector $(5; -2)$, so $B(4;\ 1)$ is on the line; check $8 + 5 \quad -13 = 0$.

**Exercise 7.2 - Calculating a projection.** Normal vector $(2;\ -1)$ ; $\lambda = \dfrac{8-3}{5} = 1$. $H = (4;3) - (2;-1) = (2;4)$. $2(2) - 4 = 0$ and $\overrightarrow{PH} = (-2;1)$ is a normal vector; distance $\sqrt{5}$.

**Exercise 7.3 - Recognising a circle.** $(x-3)^2 + (y+2)^2 = 25$; centre $(3; \quad -2)$, radius 5. For $y = -2,\ x = 3 \pm 5$ : $(-2;-2)$ and $(8;-2)$.

**Exercise 7.4 - Perpendicular bisector.** Midpoint I(1;2), normal $\overrightarrow{AB} = (4;2)$. $4(x-1) + 2(y-2) = 0$, i.e. $2x + y - 4 = 0$. I satisfies $2 + 2 \quad -4 = 0$.

**Exercise 7.5 - The best access road.** The projection H(2;4) minimises PH, with length $\sqrt{5}$ km. For every M on the road, triangle PHM is right-angled at H and $PM^2 = PH^2 + HM^2 \geq PH^2$. Equality for M=H.

**Exercise 7.6 - Parameter and set.** $(x-1)^2+(y+2)^2=5-k$. If $k<5$, circle with centre $(1;-2)$ and radius $\sqrt{5-k}$; if $k=5$, a point; if $k>5$, empty.

**Exercise 7.7 - A parabola as a locus.** $MF^2=x^2+(y-2)^2$, squared distance to the axis $y^2$. Equality: $x^2-4y+4=0$, i.e. $y=\frac{x^2}{4}+1$. Since the distances are positive, the converse holds.

**Exercise 7.8 - Delayed review.** Centre (2; 0), radius 2; $(x-2)^2+y^2=4$. For x=5, $y^2=-5$, impossible. Interpretation: the line is at distance 3 from the centre, greater than the radius.

**CQFD, exercise 18 p. 279, questions a, b, c and d.** Possible answers are $(-1;2)$, $(3;-6)$, $(\frac{1}{4};-1)$ and (1; 0). Any non-zero multiple works. For $x=4$, write $x-4=0$.

**CQFD, exercise 21 p. 279, questions a and b.** a : $x-(y-2)=0$, i.e. $x-y+2=0$. b : $4(x+1)+6(y-3)=0$, i.e. $2x+3y-7=0$. Substituting the coordinates of the points confirms membership.

## MCQ: answers and reasons

**MCQ 1.**

1. B. Coefficients of x and y.

2. A. Zero product with the normal vector.

3. D. Read the coordinates with opposite signs.

4. C. Positive square root of 9.

5. B. A sum of squares is nonnegative.

**MCQ 2.**

1. C. Same x-coordinate, zero y-coordinate.

2. A. Absolute vertical distance.

3. D. Nonzero normal vector.

4. B. Both squares must be zero.

5. C. $y^2=0$.

## True or false

1. True: through P there is a unique line perpendicular to the given line; if P lies on the line, H=P.

2. False: if c=0, a single point; if c>0, the empty set.

3. False: substituting a parametrisation of a line into the equation of a circle gives a quadratic equation with strictly positive leading coefficient, hence at most two roots.

**Differentiated teaching.** If a pupil is stuck on a technical point, suggest only the first step (domain, factor or reference sample space), then remove the help. For an independent student, ask for a second method and the handling of a boundary case. Orally, allow an initial rephrasing in the language the pupil is fluent in, then require a precise mathematical sentence in French.